CS 719 Final Project Evaluation: Content

Name	Topic
Raj Panchal	Predictive Analytics and Explainable AI for Hospital Readmission Risk

Category	**Comment**
SIGNIFICANCE / MAGNITUDE - amount of effective effort - significance of work - could this work be published?	VG The evaluation of the scope for the proposal (FAIR) stated: "The project seems to have a small scope. One dataset. A few standard methods. Novel touch is explainability, but this seems to be done in the most straightforward way, whereby the user does the analysis. In order to allow the user to contribute to explainability, the visualization interface will be required. Additional detail about the input would make the scope of the project clearer." The initial proposal was for a small-scale project for predicting one variable from one dataset. This part of the project was completed, including allowing the user to perform some explainability analysis. The project has been enhanced by adding a dashboard with four tabs (risk assessment, experiments, clinical impact, and performance). The dashboard looked good during the demo and worked smoothly – it did not crash. Some what-if analysis was performed using the dashboard. You thoroughly understand the dataset and the impact of the individual variables on the predictions. An initial comparison to LACE prediction technique was shown in the Final Presentation based on some example cases. For the Final Report, a full experimental comparison to LACE was performed and reported on. This project could lead to publishable work. Because of your careful comparison with the work of Guo et al. (which is from 2025) and your comparison with LACE (which is a real system used in hospitals), you may be able to publish work. You might aim to publish in Informatics in Medicine Unlocked. Typically, extensive surveys of related work are needed for medically related publications. Overall, this is a moderately scoped project done well.
INTRODUCTION - main idea of project - motivation	GOOD
PROBLEM SPECIFICATION - goal/objective - input and output	VG
DATA SOURCE	EX Excellent description of your data set including the distribution of 0 / 1 values.

APPROACH (METHOD) - well explained - diagram may be helpful	FAIR/GOOD Diagram was too small to read. I assume that the arrow from stage 3 to stage 7 was actually supposed to be from stage 4 to stage 5. Perhaps it could have been split into one high-level diagram matching your main colors and then three detailed diagrams. Placing the stages vertically would have allowed each one to be larger and thus more readable. The method was a standard machine learning pipeline, enhanced by using SMOTE to balance the training set and adding a dashboard at the end to increase the explainability of the results. Your explanations of the stages were informative, giving hard numbers for the work done at each stage.
RESULTS - screenshots of your software - explanation of screenshots - measures - experimental results - performance statistics	EX The results are presented clearly and in a logical order. Based on the confusion matrices, the default threshold of 0.5 seems unsuitable. Your emphasis on recall is wise. Your work on choosing a different threshold was effective and convincing. The addition of the LACE experimental results after the final presentation is a strength of the final report.
CONCLUSIONS - summarize what was done - evaluate work compared to problem statement	EX
OTHER COMMENTS	

CS 719 Final Project Evaluation: Oral Presentation

Name	Topic
Raj Panchal	Predictive Analytics and Explainable AI for Hospital Readmission Risk

Category	**Comments**				
TITLE	EX	Project name	✓	Class number & name	✓
		Your name	✓	Date	✓
PAGE NUMBERS on every slide except the title	EX				
OUTLINE AND ORGANIZATION - at most one slide for outline	GOOD				
EXAMPLE OR DEMO - appropriateness for live audience EXAMPLE: detailed input, intermediate, output DEMO: input, processing, output	EX Nice demo provided during the presentation. Thorough demo provided during the next day, which repeated the previous information and added some more. Using the dashboard, the student was able to demonstrate responses to challenging questions asked by instructor. The student thoroughly understood the input, processing, and output of the dashboard.				
READABILITY	GOOD			Legible font	✓
	Size viewable from back	-		Readable colours	✓
	The axis labels on several graphs (e.g., slides 7, 13, and 14) were unreadable from the back of the classroom.				
CREATIVE EFFORT - aesthetic design - effort made to provide pleasing, original diagrams and tables	EX The overall presentation was in a pleasing style. The many ways of organizing information, including three-part summaries, pipelines, graphs, tables, and color-coordinated results provided pleasant variety.				
DELIVERY	GOOD			Clear diction	✓
	Loudness	✓		Variation in tone	-
	Faced audience	-		Speed not too fast or slow	✓
	The student often faced the screen instead of the audience, but spoke loudly enough that he was still audible.				

ANSWERS TO QUESTIONS - addresses the question - well worded - demonstrates knowledge	EX Shubham Gandhi asked Raj Panchal: Why were certain fields selected for emphasis on the dashboard? Was it a random selection of fields? Why weren't common fields like race and gender included? Answer: Gender and race did not strongly affect the predicted results. The included fields are the ones with the greatest effect on the risk level. **This answer directly addressed the question and provided new information that would be useful to the questioner.** Dev Gohel asked Raj Panchal: Why was SMOTE used instead of undersampling or oversampling? Answer: SMOTE, which stands for Synthetic Minority Oversampling Technique, performs interpolation between two real points in the training dataset. It does not discard some data as undersampling would do. SMOTE was only used on the training dataset, not on the testing dataset. **This answer directly addressed the question and provided new information that would be useful to the questioner.** Rouzbeh Moradian asked Raj Panchal: What does "ICD-9 mapping" mean? Answer: In the raw dataset, there were three diagnosis fields named "Diagnosis 1," "Diagnosis 2," and "Diagnosis 3." To simplify, these diagnoses were reduced to values from 1 to 9 representing 9 possible clinical categories. **This answer directly addressed the question and provided new information that would be useful to the questioner.** Judith Bjorndahl asked Raj Panchal: Suppose someone is using the dashboard to compare the risk level between this software and LACE. How can they tell what features are causing the difference? Answer: The difference is because LACE does not include the number of inpatient visits. The dashboard could be changed to show that comparison explicitly. **This answer directly addressed the question and provided new information that would be useful to the questioner.**
AUDIENCE INVOLVEMENT - audience paying attention?	FAIR Medium audience interest. Five people (out of 14) were not paying attention when I sampled the audience's attention. Interest increased when the demo was given.

QUESTIONS TO OTHERS - well worded - matter of significance - demonstration of attention	OUT Raj Panchal asked Wai Paing: Will the software give a message if the receipt is of too low quality to give good results? **This question showed good understanding of the presentation.** Answer: The software will do its best. If there is a failure at any stage, there will be no output. Raj Panchal asked Niraj Asawale: If a video runs at 30 fps, would an incident lasting one second produce 30 alerts? **This question showed good understanding of the presentation. (OUTSTANDING)** Answer: There is a cool down period so that an alert is only produced every 10 seconds. I initially didn't have this and there were many alerts. The SORT algorithm tracks objects so that only one incident is alerted per object. Raj Panchal asked Samarth Deshpande: You used stacking and calibration. Why did you use both and what do each of them do in this context? **This question showed good understanding of the presentation.** Answer: There are three models that are employed to predict risk. Stacking was used to allow the meta learner to determine the weight to be assigned to each. It is not a simple average of the three results. Calibration was used as a scaling technique to adjust the numbers to a good range. Raj Panchal asked Rouzbeh Moradian: Is there a bias in your software to recommend popular movies? **This question showed good understanding of the presentation. (OUTSTANDING)** Answer: There is a bias to recommending more popular movies because they have more ratings. Raj Panchal asked Rouzbeh Moradian: Since Netflix already recommends movies based on popularity, how is your software different? How is it personalized? **This question showed good understanding of the presentation.** Answer: The recommendations are mostly for popular movies.
EXTRA DEMO - additional features that could not be demonstrated during final presentation	See EXAMPLE section above.
Mark on Presentation	**85**
Mark on Example or Demo	**86**

CS 719 Final Project Evaluation: Written Report

Name	Topic
Raj Panchal	Predictive Analytics and Explainable AI for Hospital Readmission Risk

Category		**Comment**
TITLE PAGE		EX
- project name	✓	
- your name	✓	
OUTLINE OF WRITTEN REPORT		MISSING
- is outline present?	×	
- is outline logical?	×	
BACKGROUND		FAIR In the section on previous courses, course material from CS 719 is mentioned. I wonder which course you actually mean—there was not such material in CS 719. Good, brief descriptions of Logistic Regression, Random Forest, XGBoost, and SMOTE were provided. Instead of saying “because of its known effectiveness of healthcare datasets” about XGBoost, instead you should provide evidence in terms of referenced papers for its effectiveness. The descriptions of PFI and PDP were brief but informative. The description of the LACE Index is a noteworthy part of your background information. You analyzed the strengths and weaknesses of your approach; you were supposed to analyzed the strengths and weaknesses of competing / previous approaches.
- your previous projects	-	
- relevant existing techniques	✓	
- quality of summary	-	
- analysis of strengths / weaknesses	×	
- correct format for citations in text	✓	
EXAMPLE		GOOD Very nice example showing step by step what happens, with detailed numerical values to aid comprehension. The example is relatively simple because your system is also relatively simple. At the end, there is a “live comparison” between the XGBoost conclusion and the LACE score. You say, “This shows a case where the ML model catches something that LACE misses.” You are correct that you have shown that the two methods yield opposite conclusions. However, you should argue in more detail to convince a LACE user that they indeed are making a mistake (rather than that you are).
- clarity, completeness		
- is example original?		
- appropriate for a reading audience		
- detailed input, intermediate,output		
IMPLEMENTATION		EX You gave clear explanations of three short snippets. You also explained well why these snippets were important to the project.
- prog. langs / packages / versions	✓	
- # of modules; # of lines	✓	
- show three crucial snippets of code in your written report	✓	
CONNECTION TO RECENT RESEARCH - Compare and contrast to recent research		OUT The comparison between your project and the one by Guo et al. is much improved from the comparison given in the proposal.

SUGGESTIONS FOR FUTURE RESEARCH		EX Six solid suggestions for improvement are given.
REFERENCES		GOOD As a weakness, the only research papers concerning readmissions from the last six years are Guo et al. and Huang et al. References provided for main data set and your methods. No references for software used.
- enough, up-to-date	✓	
- all in consistent, correct format	✓	
- data sets, methods, software	-	
OVERALL ORGANIZATION		VG
WRITING INTEREST: - freshness and readability - effort made to provide pleasing, original diagrams and tables		VG
WRITING QUALITY - organization of paragraphs and sentences		GOOD
Mark		**83.65**

Marking Scheme:

/10	UofR%	KEYWORD and UofR Description of Performance
10	95-100	SUPER (S): An exceptional performance.
9.5	90-94	OUTSTANDING (OUT): An outstanding performance.
9	85-89	EXCELLENT (EX): An excellent performance.
8.5	80-84	VERY GOOD (VG): A very good performance.
8	75-79	GOOD (G): A good or satisfactory performance.
7.5	71-74	FAIR (FAIR): A fairly good performance.
7	70	70-74: SATISFACTORY (SAT): A minimally acceptable performance or marginal pass.
0-6.5	0-69	0-69: FAIL (F): An unacceptable or failing performance.
0	0	0: MISSING (0): No performance.